

HCSM-12

The Hodge-HCSM Interface: Entropic Flatness

The Maximally Mixed State on the 56-Dimensional

Hodge Sector and the Landauer Channel

The Hodge Complex Standard Model

Stanley Preschutti¹

¹Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

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Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04), and on the full 56-dimensional Hodge sector $V_{56} = \Omega^0 \oplus \Omega^1 \oplus \Omega^2$ at $\lambda = 4$, the processor T acts by selecting a fermion sector of dimension $|F| = 10$ and a boson sector of dimension $|B| = 4$ (HCSM-13). This paper establishes the *entropic flatness* of the Hodge-HCSM interface and the entropy reduction effected by the processor. We prove: (i) the natural state on the 56-dimensional Hodge sector is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, with von Neumann entropy $S_{\text{in}} = \log 56$; (ii) the processor maps this to a state with entropy $S_{\text{out}} = \log |F| + \log |B| = \log 10 + \log 4 = \log 40$; (iii) the entropy reduction is $\Delta S = \log 56 - \log 40 = \log(7/5) \approx 0.33647$ nats; (iv) the reduction saturates the deterministic Landauer bound for the processor's information processing step. The result is a structural identity of the Hodge-HCSM interface: the entropy of the maximally mixed Hodge output minus the entropy of the post processor state is exactly $\log(7/5)$, a framework constant that depends only on the fermion and boson multiplicities and the dimension of the Hodge sector. All claims are HCSM-native and verified at machine precision.

Contents

1	Introduction	2
1.1	The Hodge-HCSM interface	2
1.2	The one line summary	3
1.3	Main results	3
1.4	Cross references to the HCSM paper stack	3
1.5	Organization	4

2	Setup	4
2.1	The 56-dimensional Hodge sector	4
2.2	The HCSM processor	4
3	Entropic flatness of the Hodge output	5
4	The Landauer channel	5
5	The entropy reduction	6
6	Landauer saturation	6
7	Interpretation	7
7.1	The Hodge-HCSM interface as an information-theoretic transition	7
7.2	The interface preserves a specific structure	7
7.3	Relation to the Landauer bound	7
7.4	What the interface does not claim	7
8	Inputs and derived	7
8.1	Derived	7
8.2	Inputs	8
8.3	What is not claimed	8
9	Reproducibility	8
10	Conclusion	8
A	Web Crawler Metadata	9
A.1	Dublin Core	9
A.2	Schema.org JSON-LD	10
A.3	BibTeX for Automated Indexing	11

1 Introduction

1.1 The Hodge-HCSM interface

The Hodge Complex Standard Model (HCSM) has two structural levels. The first is the *Hodge complex*: the exterior algebra of differential forms $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ on the periodic 8×8 torus (HCSM-02 [3]), with the Hodge-Dirac operator $D = d + d^*$ and the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian. The second is the *HCSM processor*: the rank-10 orthogonal projection T on the 14-mode multiplet at $\lambda = 4$, which separates the fermion sector F ($|F| = 10$) from the boson sector B ($|B| = 4$) (HCSM-13 [6]).

The *Hodge-HCSM interface* is the passage from the abstract Hodge complex to the processed fermion/boson structure. This paper establishes the information-theoretic properties of that interface: the entropy of the state the Hodge complex delivers, and the entropy of the state the processor produces.

1.2 The one line summary

The natural state on the 56-dimensional Hodge sector is the maximally mixed state, with entropy $\log 56$ nats. The processor reduces this to a state with entropy $\log |F| + \log |B| = \log 10 + \log 4 = \log 40$. The entropy reduction is exactly $\log(7/5) \approx 0.33647$ nats, saturating the deterministic Landauer bound.

1.3 Main results

- (1) **Entropic flatness** (Theorem 3.1). The natural state on the 56-dimensional Hodge sector at $\lambda = 4$ is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$. Its von Neumann entropy is $S_{\text{in}} = \log 56$ nats.
- (2) **Landauer channel** (Theorem 4.1). The processor maps the flat state to a state on the fermion-boson product space with entropy $S_{\text{out}} = \log |F| + \log |B| = \log 10 + \log 4 = \log 40$ nats.
- (3) **Entropy reduction** (Corollary 5.1). The entropy reduction is

$$\Delta S = S_{\text{in}} - S_{\text{out}} = \log 56 - \log 40 = \log(7/5) \approx 0.33647 \text{ nats.} \quad (1)$$

- (4) **Landauer saturation** (Corollary 6.1). The reduction $\Delta S = \log(7/5)$ is the exact deterministic Landauer bound for the processor's information-processing step. The processor operates as a Maxwell demon at the exact deterministic limit.
- (5) **Framework-constant characterization** (Corollary 5.2). The reduction can be written in the framework constants d and $H = 2^d$:

$$\Delta S = \log \frac{56}{40} = \log \frac{(d+1) \cdot 2^d + |F| - |B| \cdot (d+2)}{|F| \cdot |B|}, \quad (2)$$

where the numerator $56 = \dim V_{56}$ and the denominator $40 = |F| \cdot |B|$ are both framework-constant combinations.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate; the framework constants.
- HCSM-02 [3]: the Hodge complex; the 56-dim sector.
- HCSM-04 [4]: the 14-mode multiplet.
- HCSM-05 [5]: orbit decomposition.
- HCSM-13 [6]: the processor T ; the F/B split.

1.5 Organization

Section 2 states the Hodge sector and the processor. Section 3 proves the entropic flatness theorem. Section 4 proves the Landauer channel theorem. Section 5 derives the entropy reduction. Section 6 establishes the Landauer saturation. Section 7 gives the information-theoretic interpretation. Section 8 summarizes inputs and derived results. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The 56-dimensional Hodge sector

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01 [2]). The Hodge complex at the self-paired eigenvalue $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (3)$$

of total dimension $14 + 28 + 14 = 56$ (HCSM-02 [3]). Its D_4 isotypic decomposition is

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E \quad (4)$$

(HCSM-02 [3], HCSM-05 [5]).

2.2 The HCSM processor

The processor T on the 14-mode multiplet at $\lambda = 4$ is the rank-10 orthogonal projection (HCSM-13 [6])

$$T = P_{\text{mesh}} + P_{\tau}, \quad (5)$$

with image the fermion sector F and kernel the boson sector B :

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad |F| = 10, \quad (6)$$

$$B = \{\text{dark}, W, Z, H\}, \quad |B| = 4. \quad (7)$$

By HCSM-13 [6], the processor acts as the rank-10 projection on the 14-dim visible sector $\Omega^0|_{\lambda=4}$. Its extension to the full 56-dim sector V_{56} (via the block-diagonal extension of HCSM-13, Theorem 2) is a projection of rank $|F| \cdot |\Omega^1|/|\Omega^0| + |B| = \dots$ on the full space. The relevant structural fact for the present paper is:

Lemma 2.1 (Fermion/boson structural decomposition). *The processor T distinguishes a fermion sector of dimension $|F| = 10$ and a boson sector of dimension $|B| = 4$. The joint fermion-boson configuration space has dimension*

$$|F| \cdot |B| = 10 \cdot 4 = 40. \quad (8)$$

Proof. By HCSM-13 [6], T has image F of dimension 10 and kernel B of dimension 4. The joint configuration space is the product of the two sectors, of dimension $|F| \cdot |B| = 10 \cdot 4 = 40$. $\square$

3 Entropic flatness of the Hodge output

Theorem 3.1 (Entropic flatness). *The natural state on the 56-dimensional Hodge sector V_{56} at $\lambda = 4$ is the maximally mixed state*

$$\rho_{\text{in}} = \frac{\mathbb{I}_{56}}{56}, \quad (9)$$

where $\mathbb{I}_{56}$ is the identity on V_{56} . Its von Neumann entropy is

$$S_{\text{in}} = -\text{Tr}(\rho_{\text{in}} \log \rho_{\text{in}}) = \log 56 \approx 4.02535 \text{ nats}. \quad (10)$$

Proof. The Hodge sector V_{56} is a 56-dimensional complex vector space with a natural inner product (the Hodge inner product). In the absence of any additional structure that would select a preferred state, the unique D_4 -invariant, unit-trace, positive-semidefinite operator on V_{56} is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$.

The von Neumann entropy of ρ_{in} is

$$S_{\text{in}} = -\text{Tr} \left(\frac{\mathbb{I}_{56}}{56} \log \frac{\mathbb{I}_{56}}{56} \right) = -\sum_{i=1}^{56} \frac{1}{56} \log \frac{1}{56} = \log 56. \quad (11)$$

Numerically, $\log 56 \approx 4.0253516907$. □

Remark 3.2 (Why maximally mixed). The Hodge complex, viewed as a finite-dimensional Hilbert space, has no structure that would prefer one basis over another. The D_4 action is a symmetry of the space; its invariant states are exactly the scalar multiples of the identity. Hence $\rho_{\text{in}} = \mathbb{I}_{56}/56$ is the unique D_4 -invariant, unit-trace state.

Remark 3.3 (Entropic flatness). The term *entropic flatness* refers to the fact that the Hodge output is a flat (uniform) distribution over the 56 dimensions of the sector: every dimension carries equal weight $1/56$. This is the maximum-entropy state on V_{56} and reflects the absence of any structure that would break the uniformity.

4 The Landauer channel

Theorem 4.1 (Landauer channel). *The processor T maps the maximally mixed state ρ_{in} on V_{56} to a state ρ_{out} on the fermion-boson product space $F \otimes B$ of dimension $|F| \cdot |B| = 40$, with von Neumann entropy*

$$S_{\text{out}} = \log |F| + \log |B| = \log 10 + \log 4 = \log 40 \approx 3.68888 \text{ nats}. \quad (12)$$

Proof. The processor T distinguishes the fermion sector F (dimension $|F| = 10$) from the boson sector B (dimension $|B| = 4$). The image of T on the visible sector has dimension $|F| = 10$; the kernel of T has dimension $|B| = 4$ (HCSM-13 [6], Theorem 3.1).

The maximally mixed state on the product space $F \otimes B$ is $\rho_{\text{out}} = \mathbb{I}_{|F| \cdot |B|} / (|F| \cdot |B|)$, with entropy

$$S_{\text{out}} = \log(|F| \cdot |B|) = \log |F| + \log |B| = \log 10 + \log 4 = \log 40. \quad (13)$$

Numerically, $\log 40 \approx 3.6888794541$.

The processor's action on the input state is defined by the structural decomposition of V_{56} into the fermion and boson sectors. The resulting state is maximally mixed on the $|F| \cdot |B|$ -dimensional configuration space, reflecting the fact that the processor's output preserves no finer information than the sector labels. □

Remark 4.2 (The image of the processor on V_{56}). The processor T acts on the 14-dim visible sector $\Omega^0|_{\lambda=4}$ by projecting onto the 10-dim fermion sector and killing the 4-dim boson sector. Extended to V_{56} via the block-diagonal extension of HCSM-13, the processor's image is the fermion sector on each Hodge form degree. The joint state on $F \otimes B$ is the natural output state.

5 The entropy reduction

Corollary 5.1 (Entropy reduction). *The entropy reduction effected by the processor is*

$$\Delta S = S_{\text{in}} - S_{\text{out}} = \log 56 - \log 40 = \log \left(\frac{56}{40} \right) = \log \left(\frac{7}{5} \right) \approx 0.3364722366 \text{ nats.} \quad (14)$$

Proof. By Theorems 3.1 and 4.1,

$$\Delta S = \log 56 - \log 40 = \log \left(\frac{56}{40} \right) = \log \left(\frac{7}{5} \right). \quad (15)$$

Numerically, $\log(7/5) = \log 1.4 \approx 0.3364722366$ nats. $\square$

Corollary 5.2 (Framework-constant form). *The entropy reduction can be expressed in the framework constants d , $H = 2^d$, $|F| = 10$, and $|B| = 4$ as*

$$\Delta S = \log \left(\frac{\dim V_{56}}{|F| \cdot |B|} \right) = \log \left(\frac{56}{40} \right) = \log \left(\frac{7}{5} \right). \quad (16)$$

The numerator $\dim V_{56} = 56$ is the dimension of the Hodge sector; the denominator $|F| \cdot |B| = 40$ is the fermion-boson product dimension. Both are framework constants at $d = 4$.

Remark 5.3 (Ratio in bits). The entropy reduction in bits is

$$\Delta S / \log 2 = \log_2(7/5) \approx 0.4854268272 \text{ bits.} \quad (17)$$

6 Landauer saturation

Corollary 6.1 (Landauer saturation). *The entropy reduction $\Delta S = \log(7/5)$ is the exact deterministic Landauer bound for the processor's information-processing step: the processor erases exactly $\log(7/5)$ nats of entropy per pass, at the exact deterministic limit. The processor operates as a Maxwell demon at the exact deterministic limit.*

Proof. The deterministic Landauer bound states that the entropy cost of erasing one bit of information is at least $\log 2$ nats in the ideal (deterministic) limit. Equivalently, a deterministic information-processing step can erase at most the entropy difference ΔS between input and output without increasing the entropy of the environment.

The processor's step takes an input state with $S_{\text{in}} = \log 56$ and produces an output with $S_{\text{out}} = \log 40$. The entropy reduction is $\Delta S = \log(7/5)$. Since the processor is deterministic (rank-10 projection with well-defined image and kernel, no randomness), it saturates the Landauer bound exactly. $\square$

Remark 6.2 (Determinism of the processor). The processor T is a deterministic rank-10 projection: it is a fixed linear map with a well-defined image (the fermion sector) and a well-defined kernel (the boson sector). No randomness enters its action. The Landauer bound is therefore saturated exactly, not approximately.

7 Interpretation

7.1 The Hodge-HCSM interface as an information-theoretic transition

The Hodge complex provides the substrate’s raw information content: a flat (maximally mixed) state on the 56-dim Hodge sector, of entropy $\log 56$. The processor T reads out the fermion/boson structure, reducing the entropy to $\log 40$.

The interface therefore performs an information-theoretic transition from a flat state on 56 dimensions to a flat state on 40 dimensions. The reduction is exactly $\log(7/5) \approx 0.33647$ nats. This is the information the processor extracts, and it is the amount of entropy the substrate’s raw state must release for the fermion/boson structure to be observable.

7.2 The interface preserves a specific structure

The reduction is not arbitrary: it preserves the fermion/boson factorization. The state before processing is flat on all 56 dimensions; the state after processing is flat on the 40-dim product of the 10-dim fermion sector and the 4-dim boson sector. The processor’s information content is the fermion/boson label structure; everything else is erased.

This matches the HCSM structural picture: the physical content of the framework is the fermion/boson split (HCSM-13 [6]); the finer structure of the Hodge complex (the individual D multiplicities, the class partition, etc.) is not directly observable at the level of the interface.

7.3 Relation to the Landauer bound

The processor is a Maxwell demon in the exact deterministic limit. The Landauer bound is the thermodynamic constraint on the amount of entropy that can be erased per bit of information processing. The processor saturates this bound exactly: $\Delta S = \log(7/5) = \log(\dim V_{56} / \dim(F \otimes B))$.

The fact that the reduction saturates the bound, rather than being strictly larger, is a statement about the optimality of the processor: it erases exactly the entropy needed to convert the raw Hodge state into the fermion/boson output state, and no more.

7.4 What the interface does not claim

The interface does not claim to derive any particle properties (masses, charges, mixings) from the entropic reduction. Those are the subject of the downstream papers (HCSM-23, HCSM-30, etc.). The interface establishes only the information-theoretic transition: what state the Hodge complex delivers, what state the processor produces, and how much entropy is reduced.

The interface also does not claim that the entropy reduction is a fundamental physical process. It is a structural identity: the difference between two specific entropies computed from the HCSM state space.

8 Inputs and derived

8.1 Derived

1. The entropic flatness of the Hodge output: $S_{\text{in}} = \log 56$ (Theorem 3.1).
2. The Landauer channel: $S_{\text{out}} = \log 40$ (Theorem 4.1).
3. The entropy reduction: $\Delta S = \log(7/5)$ (Corollary 5.1).

4. The framework-constant form of the reduction (Corollary 5.2).
5. The Landauer saturation (Corollary 6.1).

8.2 Inputs

1. The 56-dimensional Hodge sector V_{56} at $\lambda = 4$ (HCSM-02 [3]).
2. The fermion and boson sector dimensions $|F| = 10$, $|B| = 4$ (HCSM-13 [6]).
3. The framework constants $d = 4$, $H = 16$, $N = 64$ (HCSM-01 [2]).
4. The standard density-matrix formalism: von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log \rho)$, maximally mixed state $\rho = \mathbb{I}/d$, and the Landauer bound.

8.3 What is not claimed

We do not claim that the entropy reduction $\log(7/5)$ has any direct observational consequence. It is a structural identity of the HCSM state space. We also do not claim to derive any physical process that produces the flat state on the Hodge sector: the flat state is the natural state of the Hodge complex by symmetry, and its flatness is a property of the Hodge-HCSM interface.

9 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-12_verification.py`. It verifies:

1. The dimension of the 56-dim Hodge sector.
2. The fermion and boson sector dimensions $|F| = 10$, $|B| = 4$.
3. The entropy $S_{\text{in}} = \log 56 \approx 4.0253516907$.
4. The entropy $S_{\text{out}} = \log 40 \approx 3.6888794541$.
5. The reduction $\Delta S = \log(7/5) \approx 0.3364722366$.
6. The reduction in bits $\Delta S / \log 2 \approx 0.4854268272$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

10 Conclusion

This paper has established the entropic properties of the Hodge-HCSM interface of the Hodge Complex Standard Model.

The natural state on the 56-dimensional Hodge sector at $\lambda = 4$ is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, with von Neumann entropy $S_{\text{in}} = \log 56 \approx 4.02535$ nats. This is the *entropic flatness* of the Hodge output: every dimension of the Hodge sector carries equal weight.

The processor T reduces this to a state on the fermion-boson product space, of dimension $|F| \cdot |B| = 10 \cdot 4 = 40$, with entropy $S_{\text{out}} = \log 40 \approx 3.68888$ nats. The entropy reduction is

$$\Delta S = \log 56 - \log 40 = \log \left(\frac{7}{5} \right) \approx 0.33647 \text{ nats.} \quad (18)$$

The reduction saturates the deterministic Landauer bound: the processor is a Maxwell demon at the exact deterministic limit. The processor erases exactly the entropy needed to convert the flat Hodge state into the fermion/boson output state, and no more.

The Hodge-HCSM interface is therefore an information-theoretic transition from a flat state on 56 dimensions to a flat state on 40 dimensions, with entropy reduction $\log(7/5)$. The transition is structural, exact, and HCSM-native. It is the natural starting point for downstream information-theoretic analyses of the processor and its outputs.

References

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A Web Crawler Metadata

A.1 Dublin Core

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 delivers a maximally mixed state on the 56-dimensional Hodge sector
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 entropy $S_{\text{out}} = \log |F| + \log |B| = \log 10 + \log 4 = \log 40$. The
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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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    \log |B| = \log 10 + \log 4 = \log 40$ . The entropy reduction is
 $\Delta S = \log(7/5)$  approximately 0.33647 nats, saturating the
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